Defense notes — Imagining the Target

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Target: ~20

1. Title

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Good morning, and thank you for being here. My thesis addresses a retrieval problem: finding one specific object among three quarters of a million images, when the query is an image together with a textual modification — and doing so without training any model.

2. Composed Image Retrieval

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The task is called Composed Image Retrieval. The query has two parts — here, a photograph of the Temple of Poseidon and the sentence "in an old archival photo". The correct answer is this same temple, as it appears in an archival photograph.

The two modalities are complementary: the image fixes the subject, the text specifies its transformation. Neither alone expresses the information need — in product search, "this bag, but in leather"; in an archive, "this building, at night". That composition is what makes the task interesting and hard.

3. From categories to instances

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The literature mostly studies the category-level version: for "a cartoon character on a t-shirt", any such t-shirt is correct. My thesis addresses the instance-level version: "this character on a t-shirt" — and now, of the same three results, only the Tintin shirt is correct. The retrieved object must preserve the identity of the reference under changes of medium, material, and context. Category-level methods have no mechanism for that identity constraint.

4. The i-CIR benchmark — query structure

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All experiments use i-CIR, a recent benchmark built for the instance-level task. One instance, Tintin, illustrates the structure: one reference image, five modification texts — five composed queries. The first row shows one ground truth per query — note each query actually has several, between three and fourteen here. The second row shows hard negatives, curated per instance: a statue of a different character; another character's figurine with a dog, deliberately parallel to Tintin and Snowy; a rooftop without the sign; other cartoon t-shirts. These negatives satisfy one modality but not both, so single-modality shortcuts fail by design.

5. The i-CIR benchmark — scale

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The scale: two hundred and two instances, one thousand eight hundred eighty-three composed queries — median six per instance — and a database of seven hundred fifty-two thousand images. The histograms show the distributions. The regime is needle-in-a-haystack: the positives are a vanishing fraction of the database.

6. Problem formulation — zero-shot

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Formally: given the composed query, rank the entire database by a scoring function conditioned on both the reference image and the modification text.

The central design decision is the zero-shot constraint. Supervised CIR methods — combiner networks trained on labelled triplets — cannot be used here: such triplets do not exist at the instance level, and every newly added object would require new annotation. Zero-shot alternatives from the literature, such as textual inversion, exist — but they target the category regime and do not preserve instance identity. So the entire pipeline uses frozen pre-trained models: no learned parameters, no fine-tuning, immediate generalization to unseen instances.

7. Evaluation protocol

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The system returns a ranked list; average precision is the area under that list's precision–recall curve — it is high exactly when the ground truths sit near the top. Averaging over queries gives mean average precision. The primary metric is the macro variant: queries are first averaged within each instance, then across the two hundred and two instances, so a query-rich instance cannot dominate the score. Every instance counts equally.

8. Vision–language models

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The building block for everything that follows: CLIP, a dual-encoder vision–language model. Two networks — one for images, one for text — are trained contrastively on four hundred million image–caption pairs: matching pairs are pulled together in a shared vector space, mismatched pairs pushed apart.

The consequence is the property we exploit: the relevance between any image and any sentence reduces to the cosine similarity of two vectors. The baseline backbone throughout is CLIP ViT-L/14, used frozen.

9. The BASIC method

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The benchmark's zero-shot reference method — and my baseline — is BASIC. Every database image receives two independent scores: a visual similarity to the reference image, and a textual similarity to the modification. Each branch has its own post-processing — I will return to the important steps — and the scores are fused multiplicatively. The product implements a soft logical AND: a candidate must satisfy both conditions, which is precisely what a composed query demands.

10. Score normalisation and the Harris criterion

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Two technical details of the fusion matter. First, the raw branch scores live on different scales, so each is min-normalised — the minimum is estimated once, offline — and negative residuals are clamped to zero. Second, the product is augmented with a penalty inspired by the Harris corner detector: it subtracts a term that grows when one score dominates the other. Candidates that are excellent in one modality but mediocre in the other — exactly the benchmark's curated negatives — are suppressed, just as the corner detector suppresses responses dominated by a single edge direction.

11. The structural limitation — and the proposal

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Here is the observation the thesis builds on. In BASIC, no component ever reads the reference image and the modification text together. Both branches are unimodal, so the composition — this object, under this transformation — is never modelled. "As a painting" and "next to a painting" demand different targets, yet each branch sees only its half of the query and cannot represent the difference.

The proposal: a multimodal large language model reads both query parts jointly and generates a caption of the imagined target — hence the thesis title. That caption is embedded with the same frozen text encoder and enters the fusion as a third multiplicative branch: image times text times caption.

12. Target-caption generation

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The captioner is InternVL, an eight-billion-parameter open multimodal model, used frozen. Conditioned on the reference image and the modification, it is instructed to describe the target. The instruction matters: I designed five prompting strategies — adaptive precision, constraint-priority, an anti-noise contract, a slot-fill template, and draft-and-refine — which produce five distinct captions per query, two of which you see here. The caption is embedded by the same text encoder already in the pipeline, so the zero-shot property is fully preserved.

13. Effect of the caption branch

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Does it help? On CLIP ViT-L/14, raw products, before any post-processing: the image-times-text baseline reaches 17.95 macro-mAP. Adding a single generated caption: 23.83 — plus 5.9 points, a thirty-three percent relative improvement, the largest single contribution in this work. Averaging the five instruction variants: 25.41 — and the average outperforms every individual prompt, so no prompt selection on the evaluation set is needed.

14. Post-processing I — centering

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Now the post-processing ladder, starting with centering. Embedding spaces of these models are anisotropic: all embeddings share a large common component — generic visual and linguistic statistics — which inflates every similarity score. Centering subtracts a precomputed corpus mean and re-normalises, isolating the distinctive part of each embedding. It is computed once, offline, and costs nothing at query time.

The gains are large for the image-times-text duplet — almost ten points on CLIP-L, almost twelve on SigLIP2. With the caption branch the gain shrinks, and on SigLIP2 it even dips slightly at this rung — which is informative: the caption already supplies part of the distinctive signal that centering is designed to isolate. The two mechanisms overlap; they are not simply additive.

> *If pressed on the SigLIP2 dip: −0.6 at this rung, recovered by the later rungs; the full ladder still peaks at 61.98 with centering included.*

15. Post-processing II — contextualization

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The second important step addresses the text branch. The text encoder was trained on full captions, so a terse fragment like "during sunset" is out-of-distribution. Contextualization wraps the modification in object nouns — "dog during sunset", "sunset dog" — embeds all variants, and averages them. It is the single largest post-processing gain on every backbone: over ten points for the baseline on SigLIP2, six for the caption pipeline.

16. The full ladder

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The complete picture on CLIP-L, adding one step at a time for both pipelines. The caption pipeline stays above the baseline along the entire ladder. The last two steps — semantic projection and query expansion — were tuned for raw CLIP features; once the caption branch is present they consistently hurt, so the proposed pipeline stops at contextualization. I would argue this selectivity is a finding in itself: post-processing designed around one representation does not automatically transfer to an enriched one.

17. Backbone study

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Is this a CLIP-specific effect? The same pipeline runs on four frozen dual encoders. The caption branch improves every one of them; the most striking case is SigLIP-Large, where macro-mAP doubles, from twenty-one to forty-three. SigLIP2 — the same dual-encoder family with a sigmoid contrastive objective and an improved training recipe — is the strongest base model.

18. Final results — SigLIP2

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The final configuration: SigLIP2, the five-caption average, and post-processing up to contextualization. The ladder has the same shape as on CLIP-L — peak at contextualization, projection and query expansion detrimental. The headline: 61.98 macro-mAP, fully zero-shot, against 57.69 for BASIC at its own best configuration — a gap of four point three points that persists after all of BASIC's post-processing.

19. Qualitative — success

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Qualitatively: this pink pencil case, "without sticky paper notes but filled with pens and markers". Image similarity alone retrieves lookalike cases; it cannot read. The caption resolves the negation and the required content, and both ground truths reach the top of the ranking — average precision goes from fifteen to one hundred.

20. Qualitative — failure

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And an honest failure: the mask of Agamemnon, "as a painting". The target is an abstract painting, visually remote from the reference — and every pipeline, including ours, returns real golden masks. When the modification demands a drastic domain shift, the visual branch dominates the product. This is the characteristic failure mode, and it motivates the future-work directions.

21. Limitations and future work

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Three limitations. Cost: caption generation takes about fifteen seconds per query — over ninety-nine percent of online latency — so interactive use needs smaller or quantised captioners. Fusion rigidity: a fixed product cannot down-weight the visual branch when the text demands a domain shift; query-adaptive fusion is the natural next step. And prompt sensitivity: averaging mitigates it, but does not remove it.

22. Conclusions

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To conclude. A frozen, fully zero-shot pipeline can retrieve specific object instances among seven hundred fifty-two thousand images from an image and a sentence. The core contribution is letting a multimodal model imagine the target and describe it — this supplies the joint image–text reading that dual-encoder pipelines structurally lack, gives the largest single gain, and transfers to every backbone tested. Combined with centering and contextualization — and the discipline to drop what stops helping — the system reaches 61.98 macro-mAP, four point three points above the strongest baseline configuration.

Thank you — I am happy to take questions.

Q&A map

Full CLIP-L ablation incl. paper numbers → backup 23. "The paper reports 34.35" → backup 24 (reproduction 32.48; gap in the image branch, library versions; comparisons unaffected — same environment for both methods). Latency breakdown → backup 25. The five instruction strategies with example captions → backup 26. Metric formulas are on slide 7; fusion details on slide 10.